



Research article

Modeling and performance analysis of GNSS-based train positioning system with colored petri nets

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ABSTRACT

Global Navigation Satellite System (GNSS)-based continuous and accurate train positioning is one of the key technologies for advanced train operations such as train virtual coupling. However, GNSS-based train positioning faces significant challenges in real-world scenarios due to environmental complexities and signal interferences. Considering this issue, this paper presents an approach for modeling and performance analysis of GNSS-based train positioning systems using Colored Petri Nets (CPNs). By systematically modeling the GNSS signal reception and processing process, the performance of the positioning system under various environment scenarios is evaluated. The system model integrates three types of interference signals (i.e., Amplitude Modulation (AM) signals, Frequency Modulation (FM) signals, and pulse signals) while incorporating environmental factors such as terrain obstructions and tunnel shielding. Additionally, the Extended Kalman Filter (EKF) algorithm is employed to process GNSS observation data, providing accurate train position estimations. The simulation results demonstrate that signal interferences and complex environmental conditions significantly affect the GNSS-based positioning accuracy. This study offers a comprehensive framework for evaluating the performance of GNSS-based train positioning systems in different scenarios, highlighting critical factors that influence positioning accuracy and stability.

1. Introduction

With the increasing demand for operation safety and efficiency in rail transportation, train virtual coupling has been proposed and become a hot research topic in recent years. For virtual coupling, continuous and accurate train positioning is a cornerstone. Due to the global coverage and high availability, Global Navigation Satellite Systems (GNSSs) have been widely employed for train positioning [1]. However, GNSS-based positioning faces significant challenges in complex operational environments such as mountainous regions and tunnels, where signal obstructions and interferences lead to accuracy degradation.

Extensive researches have focused on the application of GNSS in train positioning, specifically under idealized conditions with uninterrupted and unobstructed signal reception [2–4]. Existing methods, such as Extended Kalman Filters (EKF) and data fusion with inertial sensors, have been demonstrated notable in improving positioning

accuracy [5,6]. However, these works lack a comprehensive performance analysis of GNSS-based positioning in real-world operational environments such as tunnels and mountainous terrains, where signal interferences and obstructions occur frequently.

One of the critical challenges in applying GNSS-based train positioning is analyzing its performances under various operational conditions including that GNSS signals are interfered with by interference signals such as Amplitude Modulation (AM) signals, Frequency Modulation (FM) signals, and pulse signals, or obstructed by mountains and tunnels. Traditional approaches often fail to intuitively represent the dynamic interactions of GNSS signals and the environmental effects on positioning accuracy. To this end, this paper proposes a Colored Petri Net (CPN)-based approach to analyze the performance of GNSS-based train positioning systems. CPNs provide a powerful framework to describe the dynamic behaviors and concurrent processes of GNSS signal reception, signal interferences, and environmental impacts of the positioning systems. The objective of this study is to evaluate the

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performance of the GNSS-based positioning systems under varying operational scenarios including open areas, mountainous terrains, and tunnels while incorporating signal interferences and EKF data fusion for higher accuracy.

The main contributions of this paper are as follows.

- 1) A novel CPN-based modeling framework for GNSS-based train positioning systems is developed.
- 2) Dynamic influences of signal interferences (i.e., by AM signals, FM signals, pulse signals) and environmental obstructions (i.e., by mountains and tunnels) are incorporated into the model.
- 3) The performance of the GNSS-based train positioning system is quantitatively evaluated under various scenarios with the CPN model of the system.

2. Related works

Over the past decades, GNSS-based systems have been extensively applied to train positioning due to their global coverage and high accuracy under favorable conditions [1,7]. Despite these advantages, challenges such as signal interferences and terrain-induced occlusions continue to pose questions on high positioning accuracy and stability. Numerous studies have explored techniques to mitigate the impacts of signal interferences on GNSS-based positioning accuracy [8–12]. For example, Ref. [9] investigates the impact of continuous wave interference on GNSS receiver tracking performance, examining the extent of its influence by varying data bit durations or integration times. To address interference-related challenges, an interference energy detection method and a comprehensive orientation technology tailored for GNSS are introduced in Ref. [11]. Besides, a number of researches have examined the impact of environmental factors along railway lines on GNSS signal integrity. For instance, Ref. [13] explores methods of scenario-based weighted measurement noise calculation and fault detection combined with state recognition based on integrity evaluation in autonomous train positioning systems. Some studies have achieved high positioning accuracy in partially obstructed or challenging environments by fusing multi-sensor data [6,14,15].

In contrast to studies that mainly investigate GNSS signal interferences in isolation, this paper integrates environmental factors and signal interferences into the formal model of the positioning system. By utilizing model simulation, the proposed approach enables a quantitative evaluation of the system's performance. This method provides a systematic analysis and verification of the interactions among signal interferences, environmental conditions, and train positioning.

Due to the rigorous mathematical foundation, formal methods have gained recognition as effective tools for system modeling and

verification. In this work, we adopt CPNs [16] as the formal modeling framework to study the impact of signal interferences and environmental factors on GNSS-based train positioning. Petri nets are formal modeling languages that are well-known for systems characterized as concurrent, synchronous, distributed, etc. Some application examples adopting petri nets as the means of description for railway signaling systems can be found in Refs. [17–22].

3. System modeling with CPNs

Fig. 1 illustrates the modeling framework of a GNSS-based train positioning system comprising the following main parts: GNSS Receiver, Environment Scenarios, Interference Signals, GNSS Position Solution, and Evaluation Module. The module GNSS Receiver selects specific environment scenarios and receives GNSS signals. The module Environment Scenarios is composed of three submodules: Open Area Scenario, Mountain Occlusion Scenario, and Tunnel and Exit Tunnel Scenario. These submodules simulate various real-world conditions while generating GNSS signal interferences, i.e., GNSS signals are interfered with by AM, FM, and pulse signals. In the module GNSS Positioning Solution, the EKF algorithm is employed to process interfered GNSS signals for estimating the train's position. The module Evaluation compares the calculated position with the reference position and outputs the positioning error. Fig. 2 clearly illustrates the hierarchical relationships between modules of the CPN model, providing a structured view of their layered organization. The declarations of the model are categorized and summarized in two tables for clarity. Table 1 provides the colset declarations and their respective descriptions, while Table 2 provides the variable declarations and their respective descriptions.

3.1. Top-level of the positioning system model

The top-level of the CPN model is shown in Fig. 3. Each substitution transition corresponds to a specific component of the system architecture. The commands or feedbacks are represented by the tokens on the places. The GNSS Receiver processes GNSS signals transmitted by the satellite constellation. Using the EKF algorithm, the system performs a position solution to determine the train's position at the current time. The calculated position coordinates are then evaluated against reference coordinates for performance analysis. This evaluation assesses the positioning accuracy of the GNSS-based train positioning system under various environment scenarios and signal interferences.

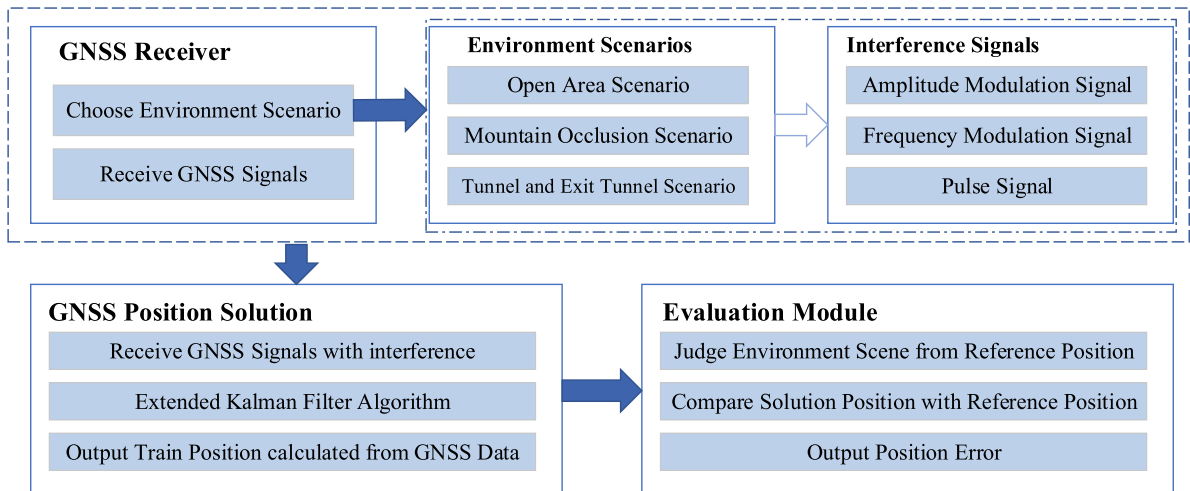


Fig. 1. Modeling framework of train positioning system considering signal interferences and environmental factors.

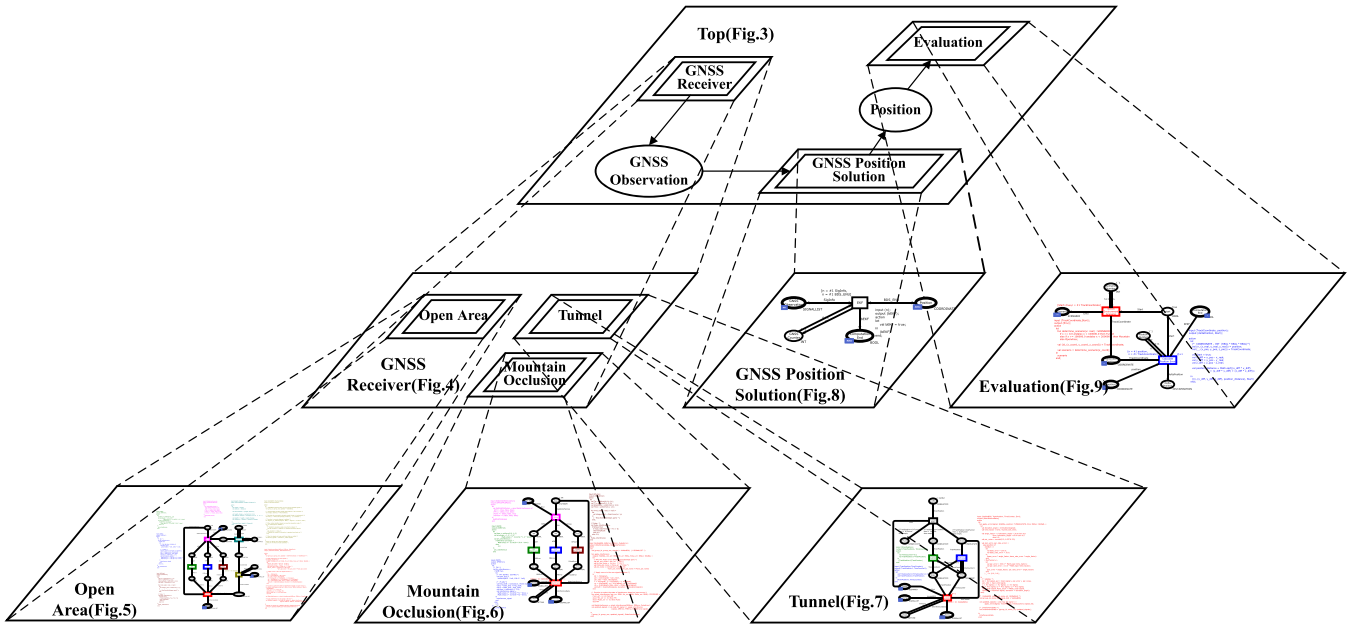


Fig. 2. Hierarchical architecture of the CPN model.

Table 1
Definitions and descriptions of colsets.

Colset definition	Description
colset BOOL = bool;	A boolean value (true/false).
colset INT = int;	An integer value.
colset INTTIME = int timed;	A timed integer value.
colset REAL = real;	A real number.
colset SIGNAL = record id: INT * psr: REAL * psr_rate: REAL * x: REAL * y: REAL * z: REAL * vx: REAL * vy: REAL * vz: REAL * clk: REAL * azimuth: REAL * elevation: REAL * Rate_clock: REAL;	Represents a signal record containing data from a single satellite. This includes pseudorange, pseudorange rate, position, velocity, and satellite metadata such as clock bias and elevation.
colset SIGNALlist = list SIGNAL;	Represents the complete set of satellite data available to the receiver at a specific moment in time. It contains a list of SIGNAL records, one for each satellite in view.
colset SIGNALLIST = product INT * INT * SIGNALlist;	Represents a comprehensive dataset for multiple consecutive time points, including all satellite data (SIGNALlist) available to the receiver, along with an associated timestamp and sequence number for each epoch.
colset SCENARIO = with OpenArea Mountain Tunnel;	Different environment scenarios.
colset STATEINFE = with AM FM Pulse Normal;	Interference states such as AM, FM, Pulse, and Normal.
colset Coordinate = product REAL * REAL * REAL;	A 3D coordinate.
colset COORDINATE = product INT * Coordinate;	A coordinate with an associated timestamp.
colset DELTAPOSITION = product INT * Coordinate * REAL;	Positional difference with an associated timestamp and error value.
colset TIMESTAMP = REAL;	A timestamp.
colset HEIGHT = REAL;	Height.
colset DISTANCE = REAL;	Distance.
colset MOUNTAIN = product HEIGHT * DISTANCE;	Mountain parameters with height and distance.
colset TUNNELSTATE = with InTunnel JustOut OutTunnel;	State of train in relation to a tunnel.
colset TUNNEL = REAL;	Tunnel-related data.

3.2. Module GNSS Receiver

The module GNSS Receiver, illustrated in Fig. 4, is designed to simulate the complete process of GNSS signal reception and processing of a train-mounted receiver. It performs the following key functions: receiving satellite signals, identifying the train's operating

environment, processing signal interferences, and producing observation results.

To simulate the behavior of GNSS signals under different environment scenarios, the model incorporates three submodules tailored for representative conditions: open areas, mountainous terrain with obstructions, and tunnels. These submodules enable realistic error

Table 2
Variable declarations and descriptions.

Variable declaration	Description
VisibleBDS, SigInfo: SIGNALLIST;	List of visible satellites.
Env: SCENARIO;	Environment scenario.
InterferenceData: INTERFERENCE;	GNSS signal data under interferences.
Mutex: MUTEX;	A mutual exclusion variable.
StateInterference: STATEINFE;	State of signal interference (e.g., AM, FM, Pulse, or Normal).
deltaPosition: DELTAPOSITION;	Error between the calculated position and the reference position.
bAM, bFM, bPulse: BOOL;	Boolean variables indicating the presence of AM, FM, or pulse interference.
AMError, FMErrror, PulseError: REAL;	Error caused by AM, FM, and Pulse interference respectively.
TrainPosition, TrainPosition1: TUNNELSTATE;	State of train's position in the tunnel scenario (e.g., InTunnel, JustOut, OutTunnel).
DelayError, TimeCounter, TimeCounter1: TIMESTAMP;	Time counters.
position, TrackCoordinate, BDS_ENU: COORDINATE;	Position of train.

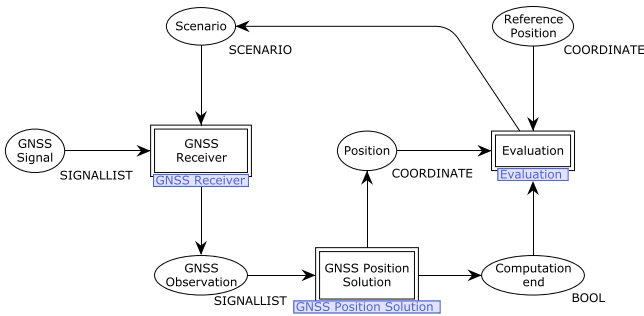


Fig. 3. Top-level of the CPN model.

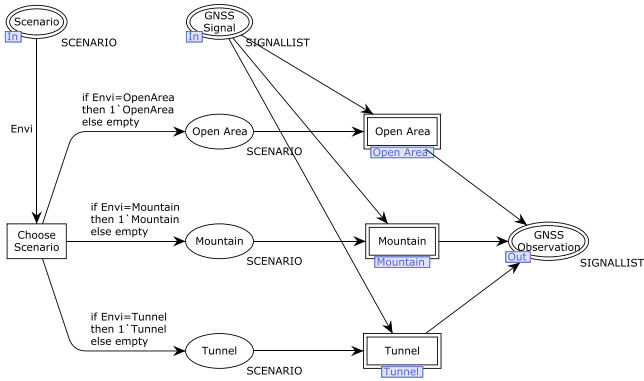


Fig. 4. Module GNSS Receiver.

modeling under different operational conditions. The transition Choose Scenario fires to select the corresponding substitute transition based on the token moving from Scenario to Choose Scenario. The tokens on the place GNSS Signal, representing the visible GNSS signals received, serve as the input data source. After processing, the output tokens move to the place of GNSS Observation.

3.2.1. Submodule Open Area

Fig. 5 illustrates the CPN model designed to simulate the impacts of GNSS signal interferences on the observation signals. This model introduces three typical types of interference signals, i.e., AM, FM, and pulse signals. Boolean variables are used to control the activation of each interference type. This design ensures that the generation process of each interference signal is both independent and easily manageable. Additionally, it accurately reflects the characteristics of realistic signal interference environments.

For the transition AM Interference, amplitude-modulated interference signals are generated by multiplying a low-frequency sinusoidal envelope (1 Hz) with a high-frequency carrier wave centered at 1575.42 MHz, corresponding to the GNSS L1 band. This configuration produces a time-varying envelope with a modulation depth of 0.5, effectively simulating realistic amplitude fluctuations observed in Radio Frequency (RF) interference sources. A similar signal interference setting is illustrated in Ref. [23]. Dynamic amplitude variations can significantly degrade GNSS signal reception, particularly in high-noise or contested environments.

The transition FM Interference simulates frequency-modulated signal interference by generating two sinusoidal components with symmetrical frequency deviations around a carrier frequency centered at 1575.42 MHz, corresponding to the GNSS L1 band. These deviations are sampled from a Gaussian distribution with a standard deviation of 75 kHz, emulating the spectral characteristics of typical FM broadcast signals [24]. This approach captures the dynamic spectral spreading commonly observed in real-world RF interference.

The transition Pulse Interference generates periodic pulse signals to simulate sudden and intense bursts of RF energy. It samples two random parameters: the pulse width T_p which represents the duration of each pulse, and the pulse interval T_i which defines the spacing between successive pulses. A uniformly distributed amplitude between 1 and 5 simulates the varying severity of the interference. At each time step, the model checks whether the current time modulo the total pulse period $T_p + T_i$ falls within the pulse's active duration. If so, an interference signal of the specified amplitude is emitted; otherwise, it remains zero. This periodic pulsed structure reflects typical burst-like interference patterns observed in GNSS environments [25], characterized by sharp, short-lived increases in signal strength followed by rapid attenuation.

The input data enter the module through the input places Open Area and GNSS Signal, representing the GNSS satellites and the initial environmental conditions. The place Bool Control and transition Mutex manage the logic for introducing signal interference, with the code in Mutex ensuring that the generation processes for different interference types do not conflict. Each interference signal is integrated into the GNSS data by firing the transition Interference. The GNSS data affected by signal interferences proceed to the observation output stage controlled by the place GNSS Number. The final results are outputted from the output place GNSS Observation.

3.2.2. Submodule Mountain

Fig. 6 illustrates the shielding effect of mountainous terrain on GNSS signals. In addition to incorporating AM, FM, and pulse signal interferences, the module introduces environmental factors of mountain height, terrain data, and obstruction distance. These elements are used

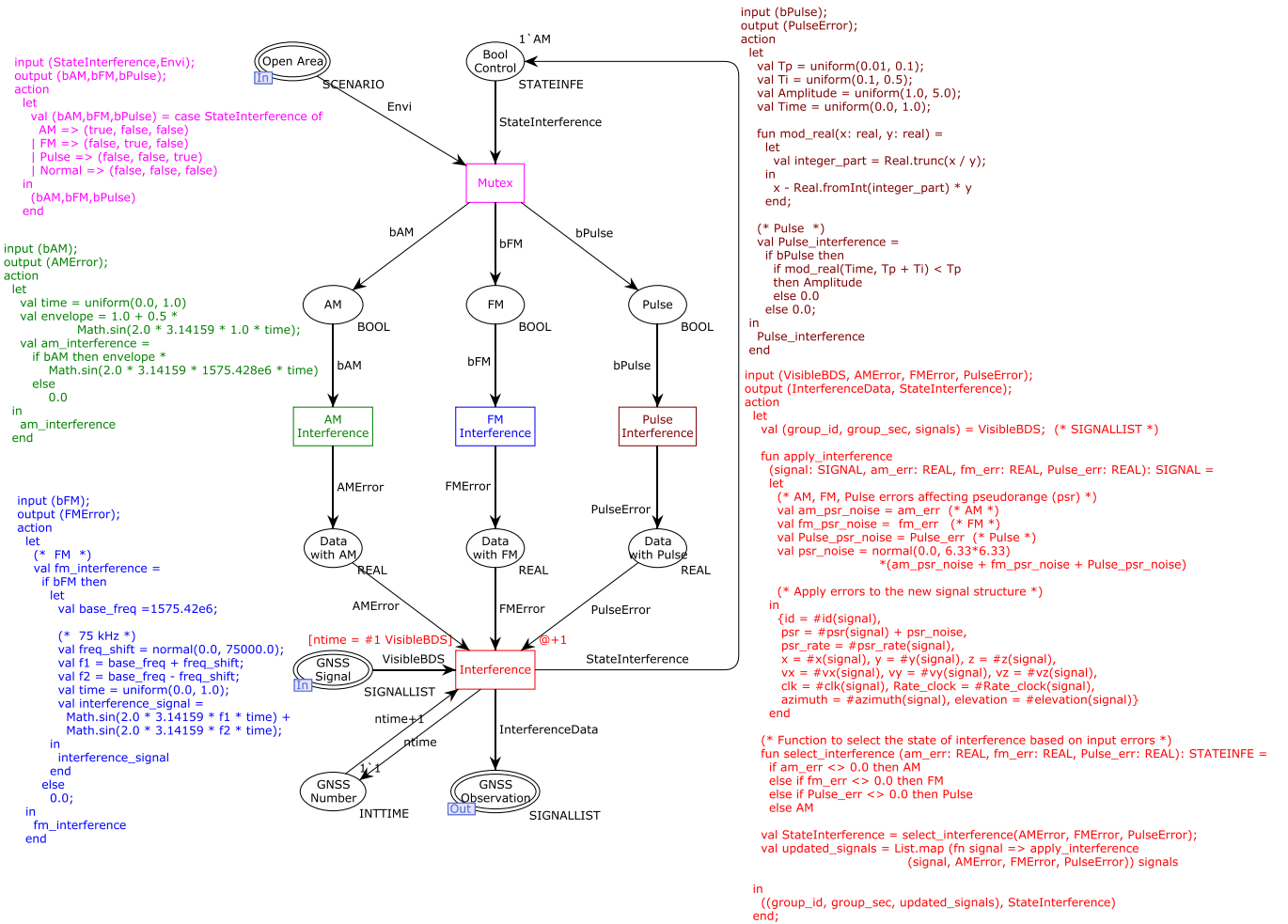


Fig. 5. Submodule Open Area.

to filter out signals from satellites that are obstructed by the terrain. The places Mountain Height and Distance store mountain elevations and spatial relationships between the terrain and GNSS signal paths. These serve as the foundational data for modeling and analyzing how terrain features obstruct satellite signals. The transition Mountain Data Fusion and its code are responsible for integrating and transmitting these data.

The place GNSS Signal stores the initial GNSS satellite data. The transition Mountain Scenario integrates mountain height and obstruction distance information with current visible satellite data to filter out obstructed signals. This process assesses satellite visibility by comparing each satellite's elevation angle with the obstruction angle derived from the mountain height and the receiver's distance to the mountain. If a satellite's elevation angle is lower than the obstruction angle, it is deemed as obstructed and excluded from the positioning calculations. After the firing of the transition Mountain Scenario, the GNSS data are stored in the place Data with Mountain Error. Additionally, visible satellite data are combined with interference signals and output to the place GNSS Observation. By incorporating a mountainous terrain scenario, this module effectively simulates the dynamic impact of terrain obstructions on GNSS signal visibility in complex environments.

3.2.3. Submodule Tunnel

Fig. 7 illustrates the process of GNSS signal reception and error variation for a train in a tunnel scenario. It focuses on the dynamic impact of the tunnel environment on the satellite signal visibility and positioning accuracy. In this module, the places InTunnel, Just Out, and Out Tunnel represent the train's current state, including three phases: inside the tunnel, just outside the tunnel, and completely outside the tunnel.

Inside the tunnel, a train is shielded by the structure blocking GNSS signal reception. During this phase, the place GNSS Signal remains empty indicating the complete obstruction of satellite signals. When the train just exits the tunnel, the GNSS signals become visible again. However, due to the complex environmental conditions near the tunnel exit, such as multipath effects, signal reflections, and delays, the GNSS observation data contain significant errors. This process is modeled by the code of the transition Tunnel Scenario, which integrates the tunnel state from the place Tunnel with visible satellite information. Over time, as the GNSS signal reception stabilizes, the positioning errors gradually decrease. This dynamic adjustment is governed by the place Time Counter. The place GNSS Observation outputs the GNSS data affected by the tunnel scenario, including both the high-error initial data and the stabilized high-accuracy data. This module effectively captures the three-phase impact of tunnel scenarios on GNSS signal reception: complete signal loss inside the tunnel, significant errors immediately after exiting, and eventual signal stabilization outside the tunnel.

3.3. Module GNSS Position Solution

Fig. 8 illustrates the process of transforming GNSS observations into train positions. GNSS observations, received from the GNSS Receiver module, are sequentially processed by the transition EKF. The completion of the EKF process is indicated by the output flag bEKF=true, which signals readiness for subsequent evaluation. Due to the limited matrix computation capability of CPN, the tokens on the place Position are generated with MATLAB. The pseudocode of the GNSS data processing workflow and position solution is presented in Algorithm 1.

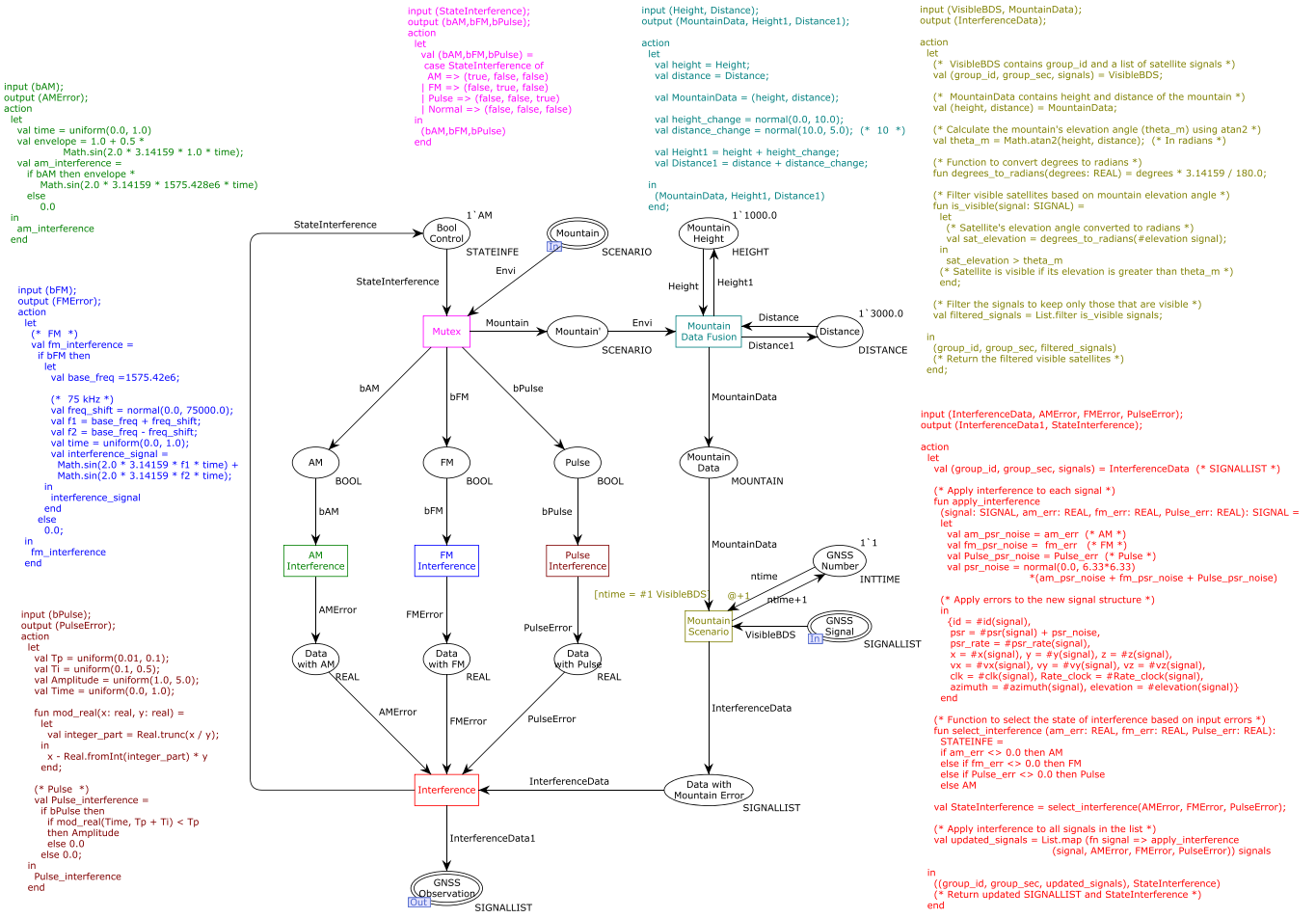


Fig. 6. Submodule Mountain.

Algorithm 1. GNSS data processing and position solution.**Input:** GNSS Observation Data**Output:** Train Position

- 1: **Initialize workspace and load libraries**
- 2: a. Set GNSS constellation flag (*GPS, BDS, combined*).
- 3: b. Load trajectory file (reference values in ECEF format).
- 4: c. Specify observation time and frequency.
- 5: d. Convert observation time to GNSS time.
- 6: **Load satellite observation data.**
- 7: **Process GNSS observation data:**
- 8: a. Initialize EKF parameters and variables.
- 9: b. Set initial position, velocity, and state variables.
- 10: **for** Each observation epoch **do**
- 11: a. Check satellite availability (minimum 4 satellites required).
- 12: b. Filter satellite data (based on elevation and SNR thresholds).
- 13: c. Compute GNSS measurements for valid satellites.
- 14: d. Estimate state variables using EKF (position, velocity, clock biases).
- 15: e. Convert and store results in ECEF, LLA, and UTM formats.
- 16: **end for**
- 17: **Save results to output file for later analysis.**

3.4. Module Evaluation

Module Evaluation in Fig. 9 is designed to dynamically assess the functionality of the train positioning system. The primary functions of the module include selecting GNSS environment scenarios, initiating GNSS data simulation, and calculating positioning errors. The module calculates the error distance between the position computed from simulated GNSS observations and the reference position by using the Euclidean distance formula in three-dimension:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad (1)$$

where (x_1, y_1, z_1) represents the reference position coordinates, (x_2, y_2, z_2) represents the calculated position coordinates, and d is the computed error distance.

The resulting error values are then output via the place Delta Position. The environment scenario is dynamically selected through the firing of the transition Select Environment Scenario. The coordinate values stored on the place Reference Position originate from an actual train trajectory, representing a segment of real-world train movement. This dataset serves as a benchmark and is precisely aligned with the GNSS signal data. The calculated positional information is input into the module via the place Position, and it is compared to the reference position data. The transition Calculate Position Error is deployed to compute the error using Eq. (1).

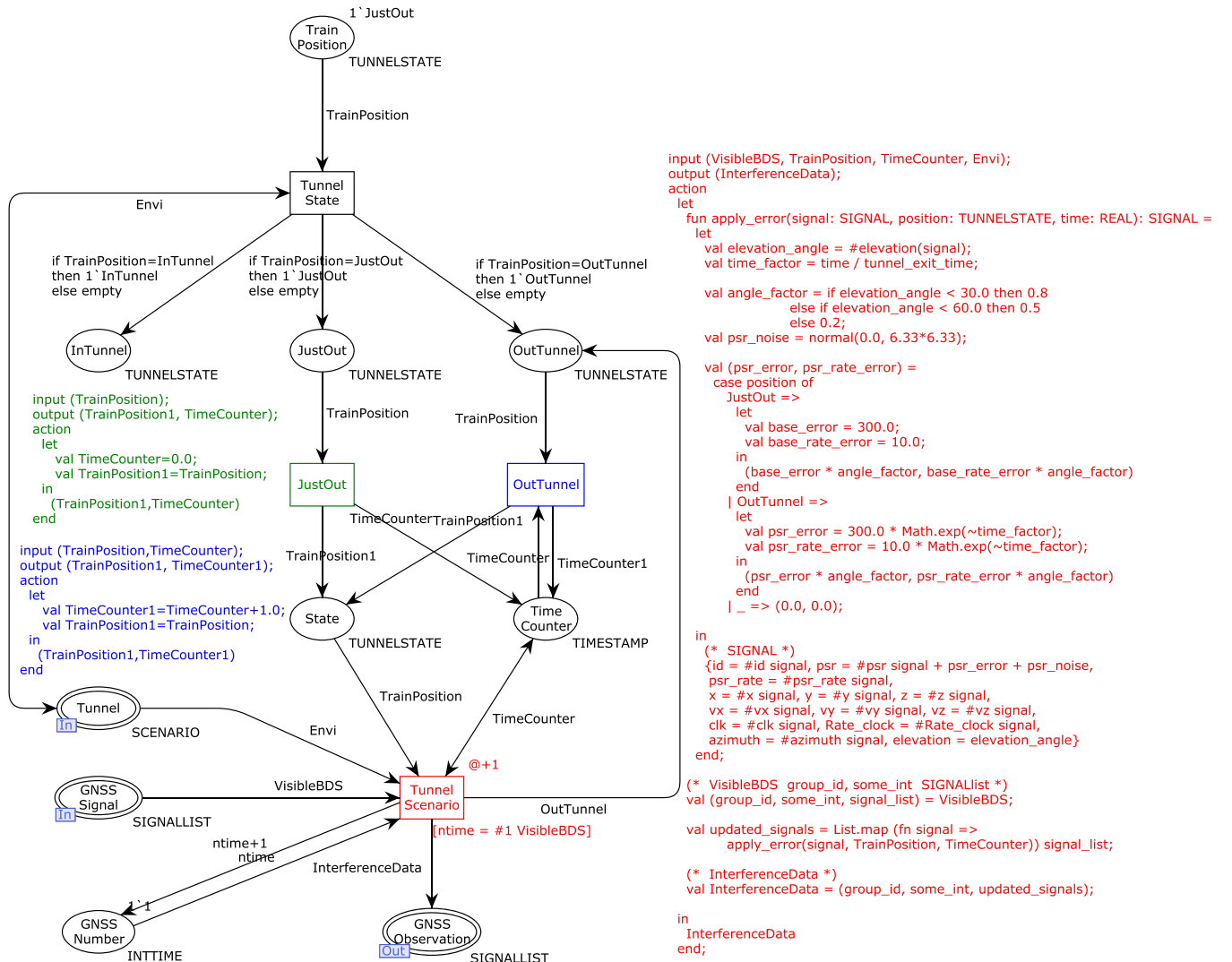


Fig. 7. Submodule Tunnel.

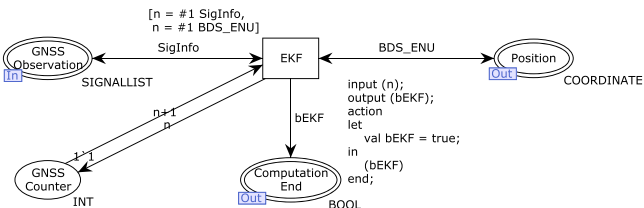


Fig. 8. Module GNSS Position Solution.

4. Simulation and analysis results

To ensure the correctness of the CPN model, formal verification is essential. Table 3 presents part of the standard state space report for the developed model. The state space is complete, comprising 5365 markings and 6410 arcs. The Strongly Connected Component (SCC) graph contains the same number of nodes and arcs, indicating the absence of cycles in the state space. The verification was performed using five representative input datasets, which cover all potential operational paths of the system. This ensures that the generated state space fully captures the range of behaviors that the model can exhibit under different input conditions. The home properties show that the initial marking is not a home marking. According to the liveness properties, the model includes 648 dead markings, but no dead or live transition instances. The fairness properties confirm that there are no infinite

occurrence sequences. These findings together demonstrate the model's soundness, boundedness, and finite execution. The existence of dead markings suggests that certain process branches reach terminal states where no further transitions are possible. This behavior is consistent with the expected functioning of the train positioning system, where a positioning cycle may conclude or the process may terminate early due to data filtering or error detection.

To simulate train positioning, loops are added to the model. Each unit of time in the model corresponds to one second in the real world. A simulation running for 600-time units represents ten minutes of station operation. To facilitate data analysis, a breakpoint monitor is set to halt the simulation when the model time exceeds 600-time units.

The dataset used for the simulation is based on a segment of the actual Jing-Shen high-speed railway. Track coordinates are derived from field measurements, while satellite ephemeris data are obtained from standard RINEX broadcast ephemeris files. At each observation epoch along the track, visible satellites and their corresponding signal parameters are calculated. Specifically, satellite positions are determined using broadcast ephemeris data through standard GNSS orbital propagation models. True pseudoranges are computed based on the geometric distance between the satellite and the receiver, assuming ideal signal propagation without any observation errors. Azimuth and elevation angles are computed by transforming the satellite coordinates into the local horizon coordinate system relative to the receiver's position. These computations adhere to established GNSS observation

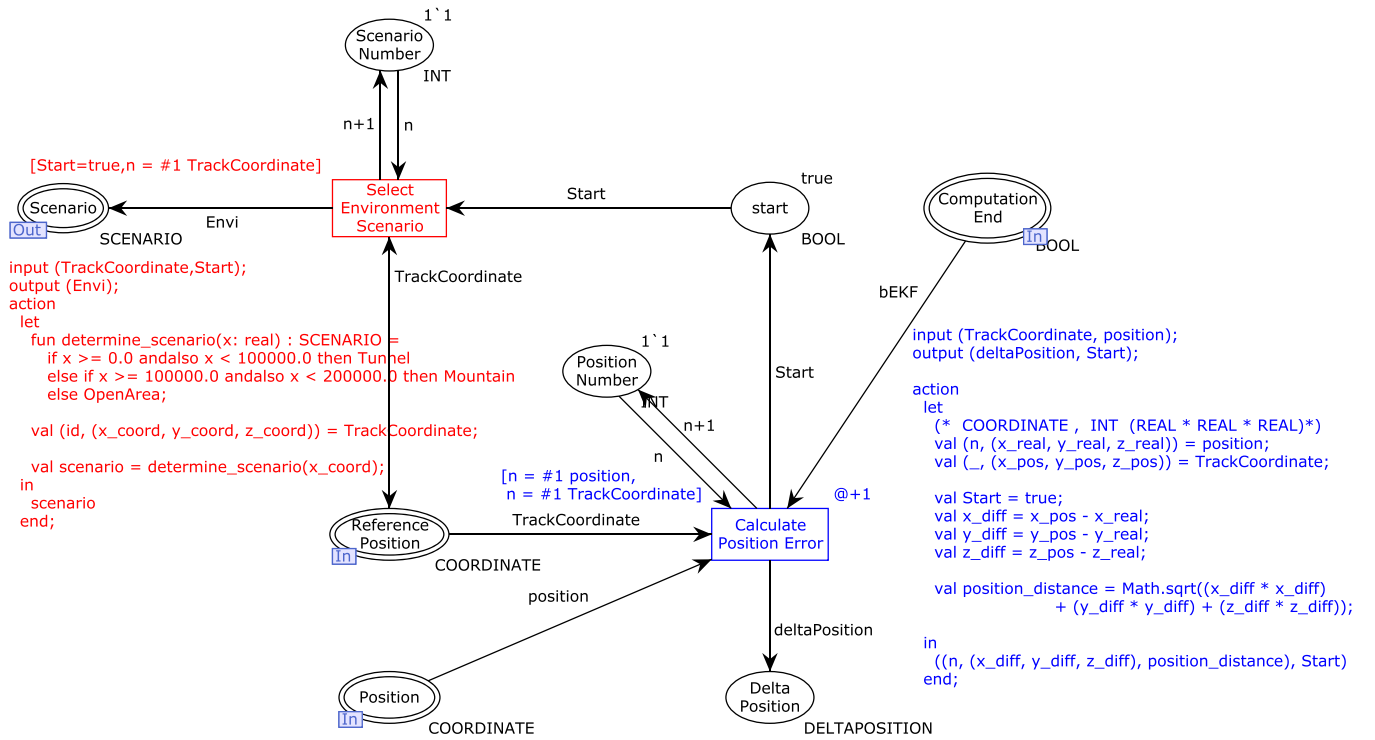


Fig. 9. Module Evaluation.

Table 3
Part of information of the standard state space report.

Statistics
State space
Nodes: 5365
Arcs: 6410
Secs: 1
Status: Full
SCC Graph
Nodes: 5365
Arcs: 6410
Secs: 0
Home properties
Home markings
None
Liveness properties
Dead markings
648 [5365, 5364, 5363, 5362, ...]
Dead transition instances
None
Live transition instances
None
Fairness properties
No infinite occurrence sequences.

modeling techniques, ensuring the reality and accuracy of the simulation inputs. Each data entry includes information for a specific observation epoch, such as ephemeris time, true pseudoranges, elevation

angles, and azimuth angles. An example of the input data is shown below.

```

1'(1, 461348, [
  {id=1, psr=23726564.3443740942,
  azimuth=168.2193480409, elevation=19.4402119310},
  {id=2, psr=23886259.8784840219,
  azimuth=296.2369811270, elevation=22.4832121883},
  .....
  {id=31, psr=21287079.5069584996,
  azimuth=228.1905278132, elevation=47.0466643097},
  {id=32, psr=21225985.1946318932,
  azimuth=34.0700281181, elevation=55.1103653523}
])
  
```

4.1. Positioning performances under different signal interference scenarios

Table 4 presents a comparative analysis of GNSS-based train positioning performances under various signal interference scenarios: interference-free condition (normal) and conditions that are interfered with by three types of interference signals: AM, FM, and pulse signals. The analysis focuses on key metrics such as mean error, standard deviation, and their respective components in the UTM coordinate system.

Table 4
Positioning performances under different signal interferences.

Scenario	Mean error/m	Standard deviation/m	Expected directional mean/m	Normalized directional mean/m	Helmert directional mean/m	Expected directional standard deviation/m	Normalized directional standard deviation/m	Helmert directional standard deviation/m
Normal	1.0279	0.0552	0.2085	0.9477	-0.2189	0.2426	0.0714	0.0776
AM	4.9484	4.0833	0.1436	1.0626	-0.2124	2.5681	2.5527	5.1820
FM	6.2241	5.2630	0.3715	1.0194	-0.0632	3.3917	3.4981	6.4434
Pulse	4.7925	3.6242	0.2009	1.1515	-0.6216	2.2564	2.4293	4.8330

Table 5

Positioning performances under different environment scenarios.

Scenario	Mean Error/m	Standard deviation/m	Expected directional mean/m	Normalized directional mean/m	Helmert directional mean/m	Expected directional standard deviation/m	Normalized directional standard deviation/m	Helmert directional standard deviation/m
Open Area	1.0279	0.0552	0.2085	0.9477	−0.2189	0.2426	0.0714	0.0776
Mountain Occlusion	1.2979	0.4528	0.2950	1.1064	−0.4783	0.2567	0.2791	0.4535
Tunnel	5.6670	6.6901	−0.1565	0.9020	−3.8520	1.5354	1.2680	7.5652

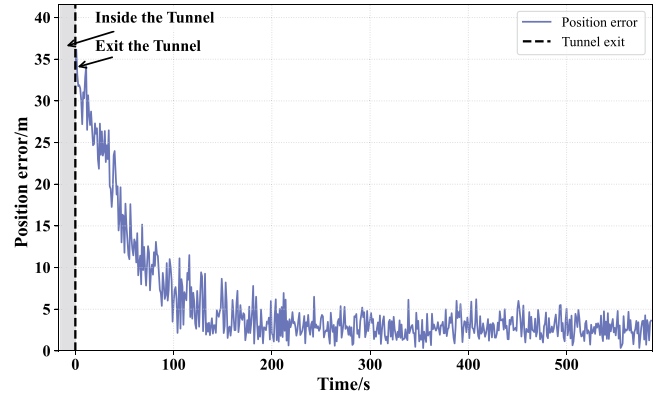
Under interference-free conditions, the positioning system demonstrates high accuracy and stability, with a mean error of 1.0279 m and a standard deviation of 0.0552 m. However, under signal interference conditions, the system's performance degrades significantly. The AM signal interference increases the mean error to 4.9484 m and the standard deviation to 4.0833 m, with the elevation direction showing the most significant fluctuations. This is likely due to the AM signal interference directly affecting signal power through amplitude variations, leading to enhanced vertical errors. The FM signal interference causes even more severe impacts, with the mean error rising to 6.2241 m and the standard deviation of 5.2630 m. The FM signal interference introduces frequency shifts exacerbating Doppler effects and significantly increasing error fluctuations in the eastward and northward directions. The pulse signal interference characterized by periodic bursts of high-intensity disruption, results in a mean error of 4.7925 m and a standard deviation of 3.6242 m. Despite a lower mean error compared to the FM signal interference, the pulse signal interference significantly affects all three spatial dimensions. This is due to its transient and repetitive nature which destabilizes signal reception and disrupts GNSS computations. These findings demonstrate that different types of signal interferences uniquely affect GNSS-based positioning accuracy. The FM signal interference exerts the most severe impact followed by pulse and AM signal interference. The elevation direction is the most sensitive to such disruptions, highlighting the direct influence of signal interference characteristics on positioning stability in different directions.

4.2. Positioning performances under different environment scenarios

Table 5 provides a detailed analysis of the performances of the positioning system under three distinct environmental scenarios: Open Area, Mountain Occlusion, and Tunnel. Key metrics of mean error, standard deviation, and UTM coordinate system components are compared to assess the accuracy and stability of the system. This comparison highlights the impact of different environmental conditions on positioning performances.

In open areas, it exhibits high precision with a mean error of 1.0279 m and minimal fluctuation, highlighting its robustness under unobstructed conditions. However, in mountainous environments, the mean error increases to 1.2979 m, reflecting the effects of signal multipath and obstructions caused by complex terrain, with the elevation direction being particularly sensitive to these factors. Tunnel scenarios show the most pronounced degradation with a mean error of 5.6670 m. This dramatic increase results from severe signal blockages in tunnels and the instability during signal reacquisition after exiting the tunnel.

Fig. 10 illustrates the evolution of position error in the tunnel environment. During the “Inside the Tunnel” phase, GNSS signals are completely blocked by the surrounding structure, resulting in a loss of positioning capability (represented by the gray shaded region). Upon exiting the tunnel (marked by the vertical dashed line), the system begins to reacquire GNSS signals; however, due to severe multipath effects and signal obstructions, the initial position errors are significantly large. As the train moves further away from the tunnel exit, the surrounding environment improves, leading to a gradual reduction in error, which eventually stabilizes. This trend highlights the critical

**Fig. 10.** Position errors in tunnel scenario.

impact of tunnel environments and their transitions on GNSS-based positioning performance.

5. Conclusion

This paper presents a CPNs-based approach to evaluate the performances of GNSS-based train positioning systems under real-world operational conditions. By constructing a hierarchical CPN model, the study successfully models the process of GNSS signal reception and interferences, as well as the impacts of complex environmental factors on the GNSS-based train positioning. The simulation results emphasize that signal interferences (i.e., interfered by AM, FM, and pulse signals) have a significant impact on the GNSS-based positioning accuracy, with FM interference exerting the most pronounced effects, particularly in the vertical direction. Environmental factors like terrain obstructions and tunnel shielding further degrade signal visibility and increase positioning errors. The model also identifies critical challenges during signal reacquisition, particularly in tunnel transitions, which exacerbate error growth.

While the proposed approach effectively demonstrates the impact of various signal interferences and environmental conditions on the performance of the GNSS-based train positioning, future research could explore the integration of additional signal interference mitigation techniques to improve the accuracy and stability of the GNSS-based positioning for railway applications.

CRediT authorship contribution statement

Siqi Wang: Writing – review & editing, Software, Data curation. **Shuting Chen:** Writing – original draft, Validation, Methodology, Formal analysis. **Daohua Wu:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Jiang Liu:** Writing – review & editing, Funding acquisition, Conceptualization.

Data Availability

Data will be made available on request.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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